

Background radiation, different half-lives and uses of nuclear radiation

Atomic structure
Paper 1 · Separate Physics only
12 questions

Cover the answers. Answer each question out loud or on paper, then check. Tick the ones you know. Learn the 3 starred (★) questions first.

	QUESTION	ANSWER	✓
1 ★	What is background radiation?	Radiation that is around us all the time, from natural and human-made sources.	☐
2	Name two natural sources of background radiation.	Rocks and cosmic rays	☐
3 ★	Name two human-made sources of background radiation.	Fallout from nuclear weapons testing and nuclear accidents	☐
4	What is radiation dose?	A measure of the risk of harm from exposure to radiation.	☐
5	Explain why an isotope with a short half-life can be very hazardous at first.	Its nuclei decay quickly, so its activity is very high.	☐
6	Give two uses of nuclear radiation in medicine.	Exploring internal organs, and controlling or destroying unwanted tissue	☐
7	Explain why alpha radiation is not suitable for exploring internal organs.	It cannot pass out of the body to be detected, and it is strongly ionising.	☐
8	State the unit of radiation dose.	Sievert (Sv)	☐
9	State how many millisieverts (mSv) equal 1 sievert (Sv).	1000 mSv	☐
10	Give two things that affect a person's background radiation dose.	Their occupation and their location	☐
11 ★	Explain why a medical tracer should have a short half-life.	Its activity falls quickly, limiting the dose to the patient.	☐
12	Explain why a medical tracer should emit gamma radiation.	It passes out of the body to be detected and is only weakly ionising.	☐